

PROBLEM STATEMENT #07 | Campus & Academic Operations

Hostel Maintenance & Issue Ticketing System

1. Problem Context & Overview

Hostel administration requires a geo-tagged ticketing pipeline to report plumbing, electrical, and carpentry issues, dispatch maintenance staff, track SLA timelines, and record resident closure sign-offs.

Target Stakeholders / Actors: *Hostel Resident, Maintenance Warden*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.

Sample Acceptance Criteria: Pass: Ticket is created, assigned a unique ID, and routed to warden queue; Fail: Incomplete form accepted without room tag.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.